

Loan Default Analysis Report

Business Intelligence Report

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Tools Used: Microsoft Excel • Power BI • GitHub

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1. Executive Summary

This project analyzes ABZ Limited's loan portfolio to identify the key factors influencing loan defaults. The objective is to uncover actionable insights that can help reduce credit risk, improve lending decisions, and strengthen portfolio performance.

An interactive Power BI dashboard was developed to monitor borrower behavior, loan characteristics, and default patterns across multiple dimensions, including loan grade, interest rate, debt-to-income ratio, employment length, and annual income.

Portfolio Overview

Metric	Value
Total Loan Applications	49,913
Total Defaulters	7,000
Default Rate	13.95%
Total Loan Amount	₦581 Million
Outstanding Principal Balance	₦35.98 Million
Average Interest Rate	14.17%

2. Business Problem

Loan default is one of the most significant risks faced by financial institutions. High default rates reduce profitability, increase credit losses, and negatively affect business sustainability.

ABZ Limited requires better visibility into borrower characteristics associated with default to strengthen credit risk management and improve lending decisions.

3. Project Objectives

The objectives of this analysis were to:

- Assess the overall performance of the loan portfolio.
 - Determine the loan default rate.
 - Identify high-risk borrower segments.
 - Analyze the relationship between borrower characteristics and loan default.
 - Develop an interactive dashboard to support business decision-making.
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4. Dataset Overview

The dataset consists of **49,913 loan applications** containing borrower information, loan characteristics, repayment status, and financial indicators.

Variables analyzed include:

- Loan Amount
 - Loan Grade
 - Interest Rate
 - Annual Income
 - Employment Length
 - Debt-to-Income Ratio
 - Outstanding Principal Balance
 - Home Ownership
 - Loan Status
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5. Data Cleaning & Preparation

Before analysis, the dataset was inspected for quality and consistency.

The following preprocessing steps were performed:

Missing Values

- Replaced missing Employment Title with "**Unknown**"
- Filled missing Loan Description values
- Standardized missing Title values
- Treated missing Annual Income values appropriately

Data Standardization

- Removed duplicate records
- Trimmed unnecessary spaces
- Standardized categorical values
- Corrected inconsistent formatting
- Verified numerical fields

Feature Engineering

Additional categories were created to support business analysis:

- Annual Income Category
- Debt-to-Income Category
- Interest Rate Category

These derived fields improved segmentation and visualization within the dashboard.

6. Dashboard Overview

The Power BI dashboard was designed as a multi-page business intelligence solution.

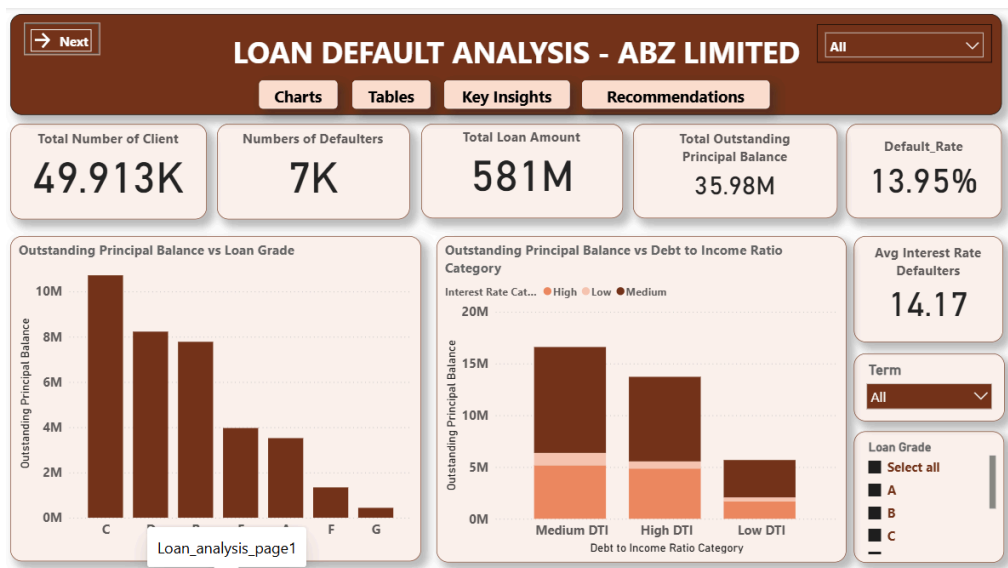
Dashboard Page 1

Portfolio Overview

This page provides:

- KPI Summary
- Outstanding Principal Balance
- Loan Grade Distribution
- Debt-to-Income Analysis
- Interactive Filters

Dashboard Page 1



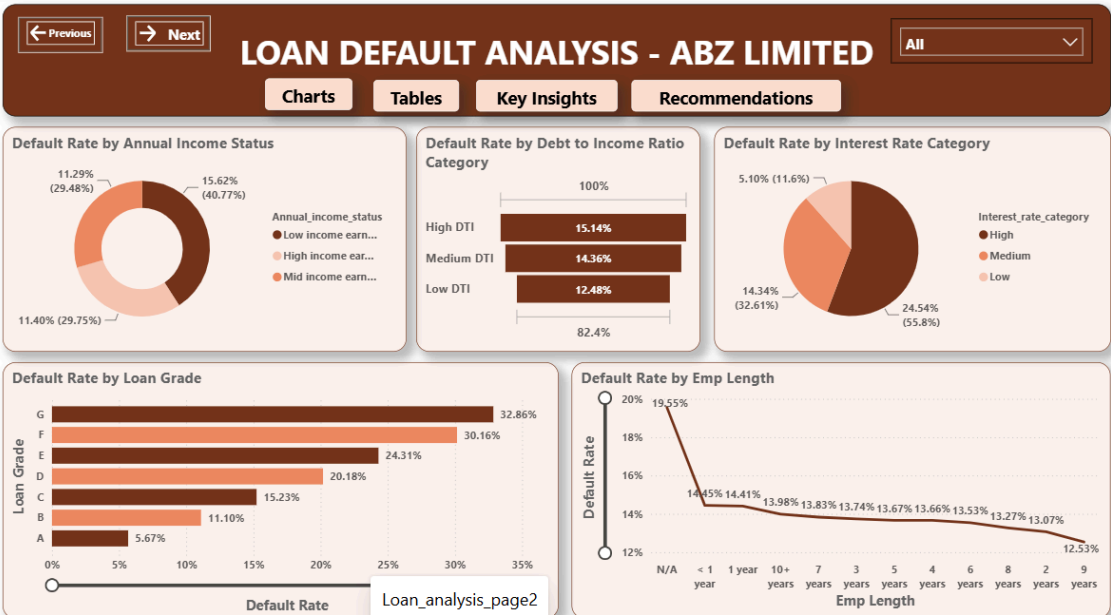
Dashboard Page 2

Risk Analysis

This page focuses on:

- Default Rate by Loan Grade
- Default Rate by Interest Rate
- Default Rate by Annual Income
- Default Rate by Employment Length
- Debt-to-Income Risk Analysis

Dashboard Page 2



7. Key Findings

7.1 Overall Portfolio Performance

The organization processed approximately **50,000 loan applications**, with **7,000 borrowers defaulting**, resulting in an overall default rate of **13.95%**.

Although the default rate appears moderate, the outstanding principal balance of **₦35.98 million** highlights significant financial exposure.

7.2 Loan Grade is the Strongest Predictor of Default

Default rates increase consistently as loan grades decline.

Loan Grade	Default Rate
A	5.67%
B	11.10%
C	15.23%
D	20.18%
E	24.31%
F	30.16%
G	32.86%

This indicates that borrowers assigned lower credit grades are substantially more likely to default.

7.3 Higher Interest Rates are Associated with Higher Default Risk

Borrowers within the **High Interest Rate** category recorded the highest default rate.

This suggests that although higher interest rates compensate for risk, they may also increase repayment challenges for borrowers.

7.4 Debt Burden Influences Default

Borrowers with **High Debt-to-Income Ratios** consistently exhibited higher default rates than borrowers with lower debt burdens.

Debt-to-income ratio should therefore remain an essential metric during loan approval.

7.5 Employment Stability Improves Repayment Performance

Applicants with missing employment history or limited work experience recorded higher default rates than borrowers with longer employment histories.

Stable employment appears to improve repayment reliability.

7.6 Outstanding Principal Balance is Concentrated Among Moderate- and High-Risk Borrowers

Most outstanding balances are concentrated within Grades **C**, **D**, and **E**, indicating greater financial exposure in these borrower segments.

8. Business Recommendations

Based on the analysis, the following recommendations are proposed:

Strengthen Credit Assessment

Apply stricter lending requirements for borrowers with lower loan grades.

Review Interest Rate Policies

Ensure that risk-based pricing remains affordable while adequately compensating for credit risk.

Introduce Debt-to-Income Thresholds

Applicants exceeding acceptable debt-to-income limits should undergo additional affordability assessments.

Improve Employment Verification

Borrowers with incomplete employment information should receive enhanced verification before loan approval.

Develop an Early Warning System

Use repayment behavior and borrower characteristics to identify customers at risk of default before delinquency occurs.

Continuous Portfolio Monitoring

Leverage interactive dashboards to monitor loan performance regularly and support data-driven decision-making.

9. Conclusion

This analysis demonstrates that loan default is strongly influenced by borrower credit quality, debt burden, employment stability, and lending conditions.

The Power BI dashboard provides decision-makers with valuable insights into portfolio performance and enables the identification of high-risk borrower segments.

Implementing the recommended measures can reduce default risk, improve portfolio quality, and support sustainable lending practices.

10. Appendix

Dashboard Pages

- Portfolio Overview

- Risk Analysis

Project Folder Structure

loan-default-analysis/

